Semester V **B. Tech CSE (AI)**

BCSAI 504: DATA MINING AND ML

3L + 0T + 0P + 3C **MM 100**

Unit 1: Introduction to Machine Learning and Data Mining

Introduction to modern data analysis, Machine Learning, supervised and unsupervised learning, Data mining definition and motivation, data mining functionalities Concept of interesting patterns, Data mining tasks, current trends, major issues and ethics in data mining, Data Mining and Knowledge Discovery in Databases.

Unit 2: Statistical Concepts & Linear Regression:

Probability Distributions, Statistical Inferences, Level of Significance, Type I and Type II Error, One Sample, Paired Samples, Independent Samples T Tests, One Way ANOVA, Chi-Square Test. Linear Regression, Scatter Plot, Correlations, R Square and Adjusted R Square, Testing of Slope, Standard Error of Estimate.

Unit 3: Logistic Regression & Decision Trees

Logit Function, Odds versus Probabilities, Nagelkerke R Square, Classification Matrix, Cut-Offs, ROC Curve. Classification & Regression Trees, Information Gain, Gain Ratio, Gini Index, Mean Squared Error, Pruning of Tree, Bagging, Random Forest, Adaptive Boosting, XG Boosting, Model Overfitting.

Unit 4: kNN& Naïve Bayes Classifiers, Association Rules Mining and Cluster Analysis

kNN as a lazy learner, Similarity Quantification, Appropriate k, Rescaling Data. Probabilistic Learning, Joint Probability, Conditional Probability, Laplace Estimator. Market Basket Analysis, Support, Confidence, Lift, Association Rules. Clustering Methods, Dendogram, Profiling of Cluster, Cluster Evaluation.

Unit 5: Dimensionality Reduction, Support Vector Machines and Text Mining

Principal Component Analysis, Eigen Values & Eigen Vectors, Singular Value Decomposition. Decision Boundaries for Support Vector Machine, Hyperplanes, Linear & Non-Linear Cases, Kernel Function, Kernel Trick, Kernel Hilbert Space, Model Evaluation. Text Data, Sentiment Analysis, Word Clouds, Term Frequency, Tokeniing.

Text Books:

1. Mitchell (2013). Machine Learning. McGraw Hill.
2. Han, Jiawei and Kamber, Micheline. (2012). Data Mining: Concepts and Techniques. Morgan Kaufman Publishers.
3. Tang, P.N., Steinback, M. and Kumar, V. (2014). Introduction to Data Mining. Pearson.

Reference Books:

1. Myatt, Glenn and Johnson, Wayne. (2009). Making Sense of Data II. Wiley.
2. Kassambara. (2017). Practical Guide to Cluster Analysis in R. STHDA
3. Silge& Robinson. (2017). Text Mining With R. SPD.